Shuyang Yu

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EDUCATION

Michigan State University East Lansing, USA
Ph.D. in Computer Science, Advisor: Dr. Jiayu Zhou Sept 2020 - June 2025
Northeastern University Shenyang, China
Bachelor of Computer Science Sept 2016 - June 2020

RESEARCH INTERESTS

Trustworthy AI: Out-of-domain generalization/detection, Robustness, Federated learning, Security, Intellectual property (IP) protection, LLMs

EXPERIENCE

Samsung Electronics America
Research scientist June 2025 -
Project: Develop machine learning methods for content-aware ads recommendation systems
Michigan State University
Graduate research assistant August 2020 - May 2025
Advisor: Dr. Jiayu Zhou
Topic: Develop robust and trustworthy machine learning algorithms across domains
GE Healthcare
Research intern May 2024 - August 2024
Advisor: Dr. Danica Xiao
Project: Enhance long-tail knowledge of LLMs for QA
JD Explore Academy
Research intern May 2021 - August 2021
Advisor: Dr. Fengxiang He
Project: Domain generalization for audio data from different domains
Data Science Lab at University of Pittsburgh
Student intern in electrical and computer engineering department July 2019 - August 2019
Advisor: Dr. Bin Gu & Dr. Heng Huang
Project: The Model Selection for Semi-supervised Support Vector Machines via Solution Path Algorithms

PUBLICATIONS

- **ConDS: Context Distribution Shift for Robust In-Context Learning.**
  Shuyang Yu, Sumyeong Ahn, Siqi Liang, Bairu Hou, Jiabao Ji, Shiyu Chang, Jiayu Zhou. 2025.
  Under review, preprint link.

- **Dynamic Uncertainty Ranking: Enhancing In-Context Learning for Long-Tail Knowledge in LLMs.**
  Shuyang Yu, Runxue Bao, Parminder Bhatia, Taha Kass-Hout, Jiayu Zhou, Cao Xiao. 2025.
  Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers) (NAACL). 2025. link.

- **Fast Online Adaptive Neural MPC via Meta-Learning.**
  Yu Mei, Xinyu Zhou, **Shuyang Yu**, Vaibhav Srivastava, Xiaobo Tan. 2025.
  The 5th Modeling, Estimation and Control Conference (MECC). 2025. link.

- **Safe and Robust Watermark Injection with a Single OoD Image.**
  Shuyang Yu, Junyuan Hong, Haobo Zhang, Haotao Wang, Zhangyang Wang, Jiayu Zhou. 2024.
  2024 International Conference on Learning Representations (ICLR). 2024. link.
- **Global Model Selection for Semi-Supervised Support Vector Machine via Solution Paths.**
  Yajing Fan*, **Shuyang Yu***, Bin Gu, Ziran Xiong, Zhou Zhai, Heng Huang, Yi Chang. 2024.
  IEEE Transactions on Neural Networks and Learning Systems (TNNLS), 2024. link.
- **Who Leaked the Model? Tracking IP Infringers in Accountable Federated Learning.**
  Shuyang Yu, Junyuan Hong, Yi Zeng, Fei Wang, Ruoxi Jia, Jiayu Zhou. 2023.
  NeurIPS 2023 Workshop on Regulatable ML. preprint link.
- **Revisiting Data-Free Knowledge Distillation with Poisoned Teachers.**
  Junyuan Hong*, Yi Zeng*, **Shuyang Yu***, Lingjuan Lyu, Ruoxi Jia, Jiayu Zhou. 2023.
  Proceedings of the 40th International Conference on Machine Learning (ICML). 2023. link.
- **Turning the Curse of Heterogeneity in Federated Learning into a Blessing for Out-of-Distribution Detection.**
  Shuyang Yu, Junyuan Hong, Haotao Wang, Zhangyang Wang, Jiayu Zhou. 2023.
  *2023 International Conference on Learning Representations (ICLR) (**Spotlight, top-25%**). 2023. link.*
- **Robust Unsupervised Domain Adaptation from a Corrupted Source.**
  Shuyang Yu, Zhuangdi Zhu, Boyang Liu, Anil Jain, Jiayu Zhou. 2022.
  2022 IEEE International Conference on Data Mining (ICDM). IEEE, 2022: 1299–1304. link.
- **Federated Adversarial Debiasing for Fair and Transferable Representations.**
  Junyuan Hong, Zhuangdi Zhu, **Shuyang Yu**, Zhangyang Wang, Hiroko H. Dodge, Jiayu Zhou. 2021.
  Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining (KDD), 617–627. link.
- **A kernel path algorithm for general parametric quadratic programming problem.**
  Bin Gu, Ziran Xiong, **Shuyang Yu**, Guansheng Zheng. 2021.
  Pattern Recognition, 2021, 116: 107941. link.
- **Tackle Balancing Constraint for Incremental Semi-Supervised Support Vector Learning.**
  Shuyang Yu*, Bin Gu*, Kunpeng Ning, Haiyan Chen, Jian Pei, Heng Huang. 2019.
  Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD), 1587–1595. link.

*equal contribution

SELECTED RESEARCH PROJECTS

Large Language Models (LLMs): in-context learning (ICL) **2023 - Present**

ICL queries the LLM by providing a set of selected examples relevant to the test query from the candidate dataset, preconditioning the models for the target task without parameter tuning.

- **Robustness against noisy samples in the candidate set of ICL:** I proposed a context distribution method to improve the quality of the candidate set used for querying the LLMs. The context distribution shift method supported different kinds of retrievers to enhance their robustness against noisy samples.
- **Enhancing ICL for Long-Tail Knowledge in LLMs:** I first investigated the limitations of existing retrieval-augmented ICL approaches for handling long-tail questions. Then I proposed a reinforcement learning-based dynamic uncertainty ranking method based on the feedback of LLMs with a budget controller to handle this challenge.

Trustworthiness: intellectual property (IP) protection **2022 - Present**

IP protection can prevent the potential financial and security impact caused by illegal reproduction or duplication of high-value models. I explored watermarking for IP protection both in a centralized and a distributed setting.

- **Watermarking without training data:** I proposed a novel watermark method based on OoD data, which fills in the gap of backdoor-based IP protection of deep models without training (ID) data. The proposed watermark method is both sample efficient and time efficient without sacrificing the model utility.
- **Tracking IP infringers:** To make the model leakage from anonymity to accountability, I proposed a watermarking method that enables ownership verification and leakage tracing at the same time for distributed system with hundreds of clients.

Trustworthiness: defense against backdoor attacks **2022 - Present**

Backdoor attacks can alter model behaviors drastically in the presence of pre-designed triggers but remains silent on clean samples.

- Uncovered the security risk of data-free knowledge distillation (KD) regarding untrusted pre-trained teacher models on diverse backdoor types distillation methods.
- Identified two potential causes for the backdoor infiltrating from the teacher to the student via data-free KD, that may inspire defense methods. To mitigate the data-free backdoor transfer, I also proposed the plug-in defensive method for data-free KD methods.

Out-of-distribution detection **2021 - Present**

Given an out-of-distribution (OoD) test sample that does not belong to any training classes, deep neural networks (DNNs) may predict it as one of the training classes with high confidence, which is doomed to be wrong. OoD detection methods can distinguish OoD samples from in-distribution (ID) samples to avoid this phenomenon.

- Proposed a novel federated OoD synthesizer to take advantage of data heterogeneity to facilitate OoD detection in federated learning (FL), allowing a client to learn external class knowledge from other non-iid federated collaborators in a privacy-aware manner.
- Bridged a critical research gap since OoD detection for FL is currently not yet well-studied in literature.

Domain adaptation **2021 - Present**

Unsupervised Domain Adaptation (UDA) provides a promising solution for learning without supervision, which transfers knowledge from relevant source domains to target domain.

- Proposed a learning paradigm that served as a general framework to integrate existing UDA approaches to improve their robustness and is powered with rigorous theoretical justifications.
- The proposed learning framework can be flexibly combined with available UDA approaches that are orthogonal to our work to improve their robustness under corrupted data.

TEACHING

Michigan State University, CSE 325: Computer Systems Spring 2021, Summer 2022, Fall 2023

- Teaching assistant for undergraduate-level courses with ~ 200 students.
- Served for grading, office hours, and Q & A sessions.

Michigan State University, CSE 847: Machine learning Spring 2022

- Volunteer teaching assistant for graduate-level machine learning courses.
- Served for grading, office hours, and Q & A sessions.

TALKS & PRESENTATIONS

- (Invited talk) Enhancing the Adaptiveness and Reliability of Machine Learning Models in Diverse Domains. *Machine Learning seminar, Samsung Electronics America*, 2025.
- (Poster presentation) Dynamic Uncertainty Ranking: Enhancing In-Context Learning for Long-Tail Knowledge in LLMs. *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (**NAACL 2025**).*
- (Poster presentation) Who Leaked the Model? Tracking IP Infringers in Accountable Federated Learning. *Michigan AI Symposium*, 2024.
- (Poster presentation) Safe and Robust Watermark Injection with a Single OoD Image. *2024 International Conference on Learning Representations (**ICLR 2024**).*
- (Poster presentation) Who Leaked the Model? Tracking IP Infringers in Accountable Federated Learning. *NeurIPS 2023 Workshop on Regulatable ML*, 2023.
- (Poster presentation) Revisiting Data-Free Knowledge Distillation with Poisoned Teachers. *Proceedings of the 40th International Conference on Machine Learning (**ICML 2023**).*
- (Spotlight presentation) Turning the Curse of Heterogeneity in Federated Learning into a Blessing for Out-of-Distribution Detection. *2023 International Conference on Learning Representations (**ICLR 2023**).*
- (Short presentation) Robust Unsupervised Domain Adaptation from a Corrupted Source. 2022 IEEE International Conference on Data Mining (**ICDM 2022**).

PROFESSIONAL ACTIVITIES

Journal/Conference Review:

- Neural Information Processing Systems (NeurIPS 2025)
- International Conference on Knowledge Discovery & Data Mining (KDD 2024)
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
- IEEE Transactions on Knowledge and Data Engineering (TKDE)
- SIAM International Conference on Data Mining (SDM)
- Neurocomputing
- ACM Transactions on Computing for Healthcare

Program Committee Member:

- FedKDD: International Joint Workshop on Federated Learning for Data Mining and Graph Analytics (FedKDD 2024).

Conference Volunteer:

- 2022 IEEE International Conference on Data Mining (ICDM), Orlando, FL.

HONORS & AWARDS

ICDM travel award	2022
Outstanding Graduate Award	2020
First Prize in China Undergraduate Mathematical Contest in Modeling (Liaoning Provincial Division)	2018

School "Excellent Student" Award	2016-2017
Second-class Scholarship	2017-2018
First-class Scholarship	2016-2017
National Scholarship (top 1%)	2016-2017